VISHWAKARMA AWARDS

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

VishwakarmaApplication ID :VK26-IITH-SE-0231

Problem Context

GUIDELINE SLIDE

Project Overview:

Project title : Smart battery Health Safety Diagnostic Device.

Team ID: VK26-IITH-SE-0231

Track: Sustainable Energy

One-line core innovation summary: An intelligent battery monitoring device that detects abnormal voltage, current, and temperature to assess battery health and provide early safety alerts.

monitors battery health, detects abnormal conditions, predicts potential failures, and provides early warnings to improve battery safety, reliability, and lifespan.

Institutional Sign-Off: Work audited and certified by Faculty Guide Prof. Sangamesh Goudappanavar, Dept. of Electrical & Electronics, Basaveshwara engineering college,Bagalkot

PROBLEMENT STATMENT:

Rechargeable batteries are widely used in electric vehicles, renewable energy storage systems, and electronic devices. However, conventional battery monitoring methods may not detect early signs of degradation or hazardous operating conditions.

Battery users and maintenance teams often lack an affordable, intelligent diagnostic solution that provides continuous health monitoring and timely warnings. Undetected overheating, abnormal voltage, excessive current, and battery degradation can lead to reduce performance, unexpected failures, costly replacements, and, in severe cases, thermal runaway or fire.

Key points:

Limited Health Visibility: Battery degradation is not always apparent during normal operation.

Overheating Risks: Temperature abnormalities may go unnoticed until the problem becomes serious.

Unexpected Failures: Battery faults can cause sudden shutdowns and operational disruptions.

High Maintenance Costs: Late fault detection can increase repair and replacement expenses.

Sustainability Challenges: Premature battery replacement increases electronic waste.

Target Beneficiaries:

Electric vehicle owners and fleet operators.

Battery manufacturers and testing laboratories.

EV service centers and battery maintenance technicians.

Solar energy storage system operators.

Industries and institutions that depend on rechargeable battery systems.

User Journey

GUIDELINE SLIDE

Visual "Before State" Requirement

Current Pain Point Mapping:

Lack of Monitoring: Battery voltage, current, and temperature are not continuously monitored.

Undetected Faults: Overheating, abnormal voltage, and battery degradation may go unnoticed.

Delayed Detection: Battery problems may be identified only after performance decreases or a fault occurs.

High Maintenance Costs: Unexpected failures can lead to expensive repairs and battery replacement.

Safety Risks: Undetected battery abnormalities may cause battery damage and hazardous situations

Quantified Inefficiency:

Time lost due to unexpected battery failures and troubleshooting.

Increased maintenance and battery replacement costs.

Reduced battery lifespan and increased battery waste.

Before State Flowchart:

Battery in Use → No Continuous Monitoring → Fault Goes Undetected → Unexpected Failure → Repair or Replacement

Visual "After State" Requirement

Future Solution Mapping:

Real-Time Monitoring: Measures battery voltage, current, and temperature continuously.

Battery Health Assessment: Estimates battery health using monitored data.

Early Fault Detection: Identifies abnormal readings and potential battery problems.

Instant Alerts: Warns users when unsafe operating conditions are detected.

Future Solution Mapping:

Real-Time Monitoring: Measures battery voltage, current, and temperature continuously.

Quantified Efficiency:

Reduced time required to identify battery abnormalities.

Faster troubleshooting through real-time monitoring and alerts.

After State Flowchart:

Battery in Use → Smart Monitoring → Fault Detection → Early Warning → Timely Maintenance

Example:Problem Context & User Journey

Problem Context

Smart Battery Health & Safety Diagnostic Device

The Challenge

Rechargeable batteries are widely used in electric vehicles, renewable energy storage systems, and electronic devices. However, battery degradation and unsafe operating conditions may go unnoticed without proper monitoring. This can lead to reduced performance, unexpected failures, higher maintenance costs, and potential safety hazards.



Key Problems

Undetected Battery Degradation
Health decreases over time, affecting performance and reliability.

Overheating & Safety Risks
Excessive temperature can damage batteries and, in severe cases, lead to fires.

Unexpected Battery Failure
Can cause sudden shutdowns and operational disruptions.

High Maintenance Costs
Late detection leads to expensive repairs or replacement.

Environmental Impact
Premature battery replacement increases electronic waste and resource consumption.

The Need

An intelligent, affordable, and reliable battery diagnostic solution that continuously monitors battery health, detects abnormal conditions, and provides early safety warnings.

User Journey

Smart Battery Health & Safety Diagnostic Device

Before Using the Device

(Current Pain Points)

No Continuous Monitoring
Battery voltage, current, and temperature are not monitored regularly.

Undetected Faults
Overheating, abnormal voltage, and degradation may go unnoticed.

Delayed Detection
Problems are identified only after performance decreases or failure occurs.

High Maintenance Costs
Unexpected failures lead to expensive repairs or replacement.

Safety Risks
Undetected abnormalities may cause battery damage and hazardous situations.

Before State Flow



After Using the Device

(Future Solution)

Real-Time Monitoring
Measures battery voltage, current and temperature.

Battery Health Assessment
Estimates battery health using monitored data.

Early Fault Detection
Identifies abnormal readings and potential problems.

Instant Alerts
Warns users when unsafe conditions are detected.

Preventive Maintenance
Helps users take timely action to reduce damage and unexpected failures.

After State Flow



The proposed device transforms battery maintenance from reactive fault handling into proactive health monitoring, helping improve safety, reliability, and sustainability.

AI Generated

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System Architecture & Specs (12 Slides)

GUIDELINE SLIDE

System Block Diagram:

Battery → Voltage, Current & Temperature Sensors → Microcontroller → Battery Health Analysis → Display & Alert System

System Components:

Battery: The battery being monitored and diagnosed.

Sensors: Measure battery voltage, current, and temperature.

Microcontroller: Processes sensor data and checks for abnormal conditions.

Diagnostic Algorithm: Analyzes readings to estimate battery health and detect potential faults.

Display Unit: Displays battery parameters and health status.

Alert System: Provides warnings when abnormal conditions are detected.

Working Principle:

The sensors collect battery data and send it to the microcontroller. The microcontroller analyzes the readings, identifies abnormal conditions, and displays battery status. If unsafe conditions are detected, the system activates an alert.

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Target specifications:

Operational Targets:
Voltage Monitoring: According to the selected battery's voltage range.
Current Monitoring: Measurement of charging and discharging current.
Temperature Monitoring: Detection of abnormal battery temperature.
Health Assessment: Estimation of battery State of Health (SoH).
Fault Detection: Identification of abnormal voltage, current, and temperature.
Response Time: Near real-time monitoring and alerts.
Power Supply: Compatible with the selected battery system or a separate regulated supply.
User Interface: Digital display for readings and battery status.

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Hardware Selection

Proposed Hardware Components:
Microcontroller: ESP32 development board.
Voltage Sensor Voltage divider circuit or suitable voltage sensor module.
Current Sensor: ACS712 current sensor or a suitably rated alternative.
Temperature Sensor: DS18B20 digital temperature sensor.
Display: 16×2 LCD with I²C interface.
Alert System: Buzzer and LED indicators.
Power Supply: Regulated power supply suitable for the selected hardware.
Communication Wi-Fi or Bluetooth using ESP32, if wireless monitoring is required.

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CAD Design & MechanicalSpecs

Smart Battery Health & Safety Diagnostic Device

CAD Design | 3D Renders | Mechanical Specifications

Smarter Monitoring • Safer Batteries • Greener Future

1. 3D Isometric View (Complete Assembly)



2. Exploded Assembly View



Assembly Parts List

No.	Component	Description
1	Top Enclosure	ABS Plastic (with display cutout & ventilation)
2	LCD Display	16x2 LCD with I²C
3	ESP32 Microcontroller Board	Data processing & communication
4	Voltage Sensor	Voltage divider module (0–25V)
5	Current Sensor	ACS712 (0–30A)
6	Temperature Sensor	DS18B20 (waterproof)
7	Buzzer	Audio alert
8	LED Indicators	Power / Warning / Fault
9	Battery Terminal Board	Input connectors (BAT+ / BAT−)
10	Bottom Enclosure	ABS Plastic (base)

Mechanical Specifications

Overall Dimensions (L × W × H)	150 × 100 × 40 mm
Enclosure Material	ABS Plastic (Flame Retardant)
Enclosure Weight	≈ 180 g (empty)
Mounting	Screw holes / Wall mount (optional)
Display Window	55 × 16 mm (LCD cutout)
Connector Openings	Battery terminals, USB-C, sensor ports
Ventilation	Side slots for heat dissipation
Operating Temperature	0°C – 50°C
Protection Class	IP20 (indoor use)

Material Specifications

Part	Material / Grade
Enclosure	ABS (UL94 V-0)
PCB	FR-4 (1.6 mm)
LCD Window	Acrylic / Polycarbonate
Connectors	Terminal Block (3.5 mm)
Screws	Stainless Steel (M2 × 6)

3. Orthographic Views & Dimensions



4. Internal Component Layout



5. 3D CAD Renders (Different Angles)



Key Features of the Design

- Compact and portable design
- Universal use (supports different battery types)
- Modular internal layout for easy maintenance
- Proper ventilation for heat dissipation
- Clear display and status indicators
- Safe and reliable sensor integration

A well-designed hardware system ensures accurate monitoring, reliable performance, and long-term durability.

Smart Technology | Safer Batteries | Sustainable Energy

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Example: CAD Design & Mechanical Specs



ESP32 Microcontroller: Processes sensor data and controls the device.

Voltage Sensor Measures battery voltage and detects abnormal voltage levels.

Current Sensor (INA226): Measures battery charging and discharging current.

Temperature Sensor (DS18B20): Monitors battery temperature and detects overheating.

LCD Display (16×2): Displays battery voltage, current, temperature, and status.

Buzzer and LED Indicators: Provide audible and visual alerts when abnormal conditions are detected.

Protective Enclosure (ABS Plastic): Houses and protects the electronic components while providing mounting points for the display, connectors, and PCB.

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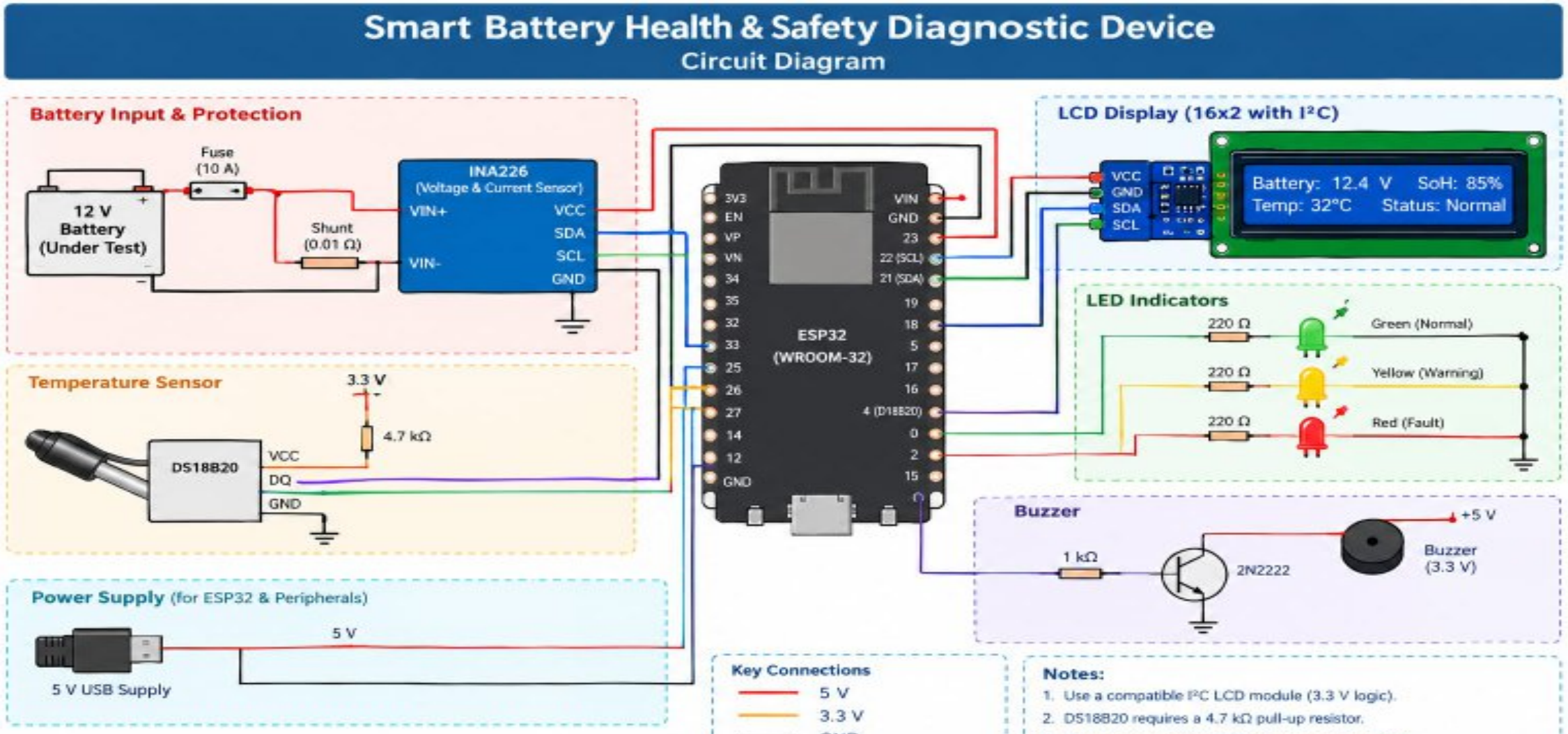
Electronics Schematics & BOM (12 Slides)

Component:	Qty:	Estimated cost:
ESP32 board	1	₹400
INA226 module	1	₹250
DS18B20 sensor	1	₹100
LCD display	1	₹200
LEDs and buzzer	1set	₹50
Resistors and driver	1set	₹30
Fuse and holder	1set	₹80
Power supply	1	₹150
Wires and PCB	1set	₹150
ABS enclosure	1	₹250
Connectors and screws	1set	₹100
Total		₹1760

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Example: Electronics Schematics & BOM



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Example: Maker Journey





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Safety & Hazard Mitigation (1 Slide)

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Safety & Hazard Mtigation

1. Safety & Hazard Mitigation
Physical Fail-Safes:
Use a fuse and reverse-polarity protection.
Monitor battery temperature to detect overheating.
Provide a protective enclosure and insulated wiring.
FMEA (Failure Mode and Effects Analysis):
Sensor Failure: Calibrate sensors and detect abnormal readings.
Power Failure: Provide a stable power supply and low-voltage alerts.
Overheating: Trigger buzzer and LED warnings when temperature exceeds safe limits.
Short Circuit: Use appropriate fuses and protected connections.

Standards Compliance

2. Standards Compliance

IEC 62133: Relevant battery safety requirements for portable sealed batteries.

IEC 61010-1: Safety considerations for electrical measurement equipment.

IP Rating: Select an enclosure with a suitable ingress-protection rating.

Electrical Safety: Use properly rated components, insulated connections, and appropriate protection circuits

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Roadmap

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4-Month Fabrication Plan

Month 1: Design & Component Selection

Finalize circuit design and system architecture.

Select sensors, ESP32, display, and safety components.

Month 2: Hardware Development

Assemble the circuit and connect sensors.

Develop the battery monitoring system.

Month 3: Software Integration & Testing

Program the ESP32 for real-time monitoring.

Test voltage, current, temperature, and alert functions.

Month 4: Prototype Validation

Calibrate sensors and verify measurement accuracy.

Test safety alerts and improve reliability.

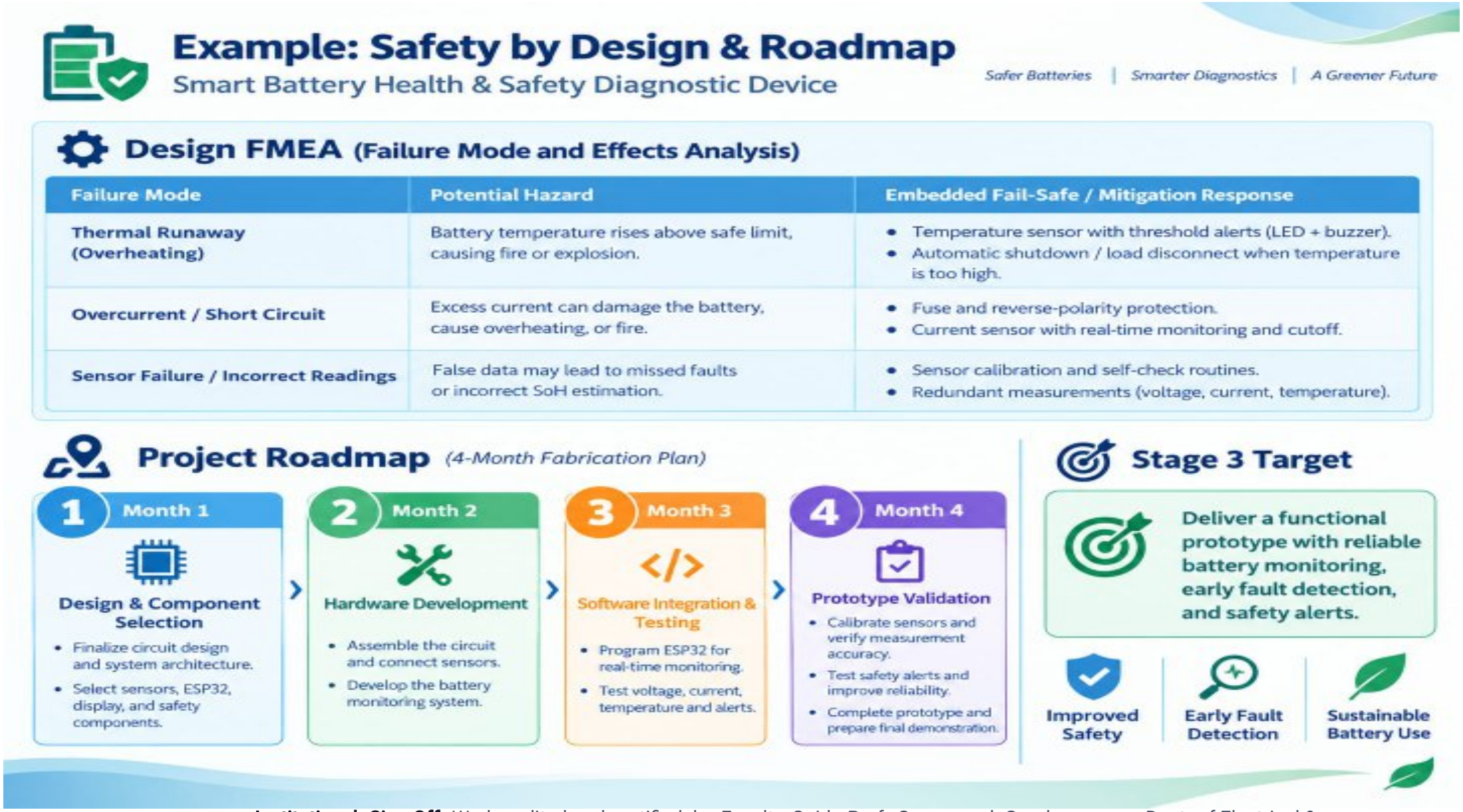
Complete the prototype and prepare the final demonstration .

Stage 3 Target:

Deliver a functional prototype with reliable battery monitoring, early fault detection, and safety alerts.

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Example: Safety by Design & Roadmap



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Technical Challenges& Risks

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Engineering Bottlenecks

Sensor Accuracy: Maintaining accurate voltage, current, and temperature measurements.

Battery Compatibility: Supporting different battery types and voltage ranges.

Heat Management: Preventing overheating during continuous monitoring.

Health Estimation: Accurately estimating battery State of Health (SOH).

Power Consumption: Minimizing the device's power usage.

System Integration: Ensuring smooth communication between sensors, ESP32, and display.

Potential Solutions & Risk Mitigation:

GUIDELINE SLIDE

1.Potential Solutions

Real-Time Monitoring: Use voltage, current, and temperature sensors to continuously monitor battery conditions.

Early Fault Detection: Implement an ESP32-based system to detect abnormal readings.

Battery Health Estimation: Use battery-specific algorithms to estimate State of Health (SoH).

Instant Alerts: Use LEDs, a buzzer, and a display to notify users of potential problems.

Universal Compatibility: Use modular sensors and configurable settings to support different battery types.

2. Risk Mtigation

Overheating: Set temperature limits and trigger early warnings.

Electrical Damage: Use fuses, reverse-polarity protection, and correctly rated components.

Incorrect Readings: Calibrate sensors and validate measurements.

System Failure: Test the prototype thoroughly and use reliable connections.

Battery Hazards: Start with low-voltage testing and retain the battery's existing protection system.

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Example: Technical Challenges & Risks

1. Technical Challenges
Sensor Accuracy: Measuring battery voltage, current, and temperature accurately.
Battery Compatibility: Supporting different battery types and voltage ranges.
Battery Health Estimation: Accurately estimating battery health using sensor data.
Real-Time Monitoring: Detecting abnormal conditions and generating alerts quickly.
Power Management: Ensuring stable power supply to all electronic components.
2. Risks and Mitigation
Overheating Risk: Use temperature monitoring and appropriate warning thresholds.
Electrical Damage: Include fuses, reverse-polarity protection, and properly rated components.
Incorrect Readings: Calibrate sensors and validate measurements.
System Failure: Test the device thoroughly and use reliable connections.
Battery Safety: Begin testing with a low-voltage, current-limited setup and never bypass the battery's existing protection system.

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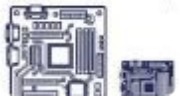
References & Sources

GUIDELINE SLIDE

(Mention the References and Sources used in your PPT)



Source: www.autodesk.com



https://static.vecteezy.com/system/resources/previews/073/726/958/non_2x/motherboard-design-with-electronic-components-and-circuitry-outline-depicti n

Source: janetmedicalclinic.com

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